Shivram S

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.75
Kendriya Vidyalaya No. 2, Kochi	Class XII (Central Board of Secondary Education (CBSE))	2024	99.40%
Kendriya Vidyalaya No. 2, Kochi	Class X (Central Board of Secondary Education (CBSE))	2022	98.60%

ACHIEVEMENTS

- **International Collegiate Programming Contest (ICPC)**
- Asia West Continental Finalist 2025-26 - Rank 71 among top 100 teams from India, Pakistan, Afghanistan and Iran.
- Chennai Regionals 2025-26 - Rank 25 among 100+ Indian teams.
- Amritapuri Regionals 2025-26 - Rank 42 among 300+ Indian teams.
- All India Prelims 2025-26 - Rank 8 among 2200+ Indian teams.

- **IICPC Quantfest 2025**: Rank 171 out of 1750 participants.

- **IICPC Codefest 2026**: Rank 705 out of 13,000+ participants.

- **Expert** on **Codeforces** with a peak rating of 1655 (username: shivrm).

- Received the **Academic Excellence Award** from IIT Hyderabad for outstanding academic performance.

- Was among the **top 35 students** out of 44000+ candidates in the **Indian National Chemistry Olympiad (INChO)** 2024, and was selected for the Orientation-cum-Selection Camp for the International Chemistry Olympiad (IChO).

- Placed among national top 1% of candidates out of 16000+ participants in the **National Standard Examination in Astronomy (NSEA)** and state top 1% of candidates in the **National Standard Examination in Physics (NSEP).**

EXPERIENCE

Research Project -- Approximate Nearest Neighbors Search

- Worked on the parallelization of graph-based ANN search algorithms using **CUDA** under Prof. Jyothi Vedurada at IITH.
- Refactored and extended the **Vamana** graph build algorithm to handle datasets containing millions of points.
- Used techniques such as kernel fusion and memory coalescing to achieve significant speedups over the baseline.
- Implemented and optimized **workload generation** for dynamic ANNS systems in C++.
- Built a parallel implementation of the **Chamfer distance metric** for computing distances between point clouds.

AI/ML Intern -- FinStocks AI

- Developed a **backtesting engine** for training and testing models on candlestick data from the Indian equity market.
- Built a news pipeline that ingests news from multiple sources and does **sentiment analysis** using reasoning models.
- Added relevance-based news filtering to the pipeline by training a **text classification model**.
- Implemented a trading strategy selector by using k-Nearest Neighbors to identify past days with similar trends.
- Built an **agent loop** that iteratively generated trading strategies through web scraping and automated backtesting.

PROJECTS

SVIFT: Peer-to-Peer File Sharing Application | GitHub

- Used **Rust** and **Tokio** to build a P2P file transfer system that can concurrently manage transfer of multiple files.
- Implemented a peer tracker in **Axum**, as well as trackerless peer discovery through peer exchange and local multicast.
- Added a TUI frontend to the application using **Crossterm** and **Ratatui** for easily managing transferred files.

Ocular: 3D Rendering Engine | GitHub

- Implemented a ray-tracing 3D renderer in **Rust** that simulates light-ray physics to render 3D scenes.
- Implemented a real-time 3D renderer in **C** that uses camera projection transformation to render triangle meshes.
- Achieved 5x speedup by using **native threads** and **OpenMP** to distribute rendering across multiple CPU cores.

RISC-V CPU and Cache Simulator | GitHub

- Built a RISC-V **assembler** and **CPU simulator** in **C** with an easy-to-use CLI interface.
- Added **debugging tools** such as register and memory dumps, breakpoints, stack traces and step-by-step execution.
- Implemented a CPU **data cache simulator** with support for caches with various replacement policies.

RISP: Lisp-like Programming Language | GitHub

- Implemented an efficient lexer and recursive-descent parser in **Rust** to parse a LISP-like grammar.
- Created a **macro system** to implement core language features such as conditionals and loops using macros.
- Built an **interpreter** for the language, along with a **standard library** for supporting I/O.

Unherd: Debiased Recommender System | GitHub

- Used **Python** and **PyTorch** to develop a recommender system that mitigates herding effects due to popularity bias.
- Implemented and trained base recommender models such as **matrix factorization** and **LightGCN** and applied **counterfactual methods** to estimate popularity bias of each recommendation and compute true merit.
- Implemented reranking methods such as **xQuAD** and **calibrated popularity** to boost recommendation diversity.

RELEVANT COURSES

Computer Science: Computer Architecture, Compilers-II, DBMS-I, Operating Systems-I, Data Structures and Applications, Algorithms (A+), Theory of Computation, Introduction to Programming (A+), Discrete Mathematics

Artificial Intelligence: Deep Learning (A+), Foundations of Machine Learning (A+), Optimization (I and II)

Mathematics: Calculus, Differential Equations, Numerical Analysis, Probability, Linear Systems and Signal Processing

SKILLS

Programming Languages: Python, C/C++, Rust, CUDA, SQL, JavaScript

AI/ML: PyTorch, Pandas, NumPy, scikit-learn, Matplotlib, LangChain,

Tools: Docker, Git/GitHub, Linux, PostgreSQL, MySQL, OpenMP, Svelte

POSITIONS OF RESPONSIBILITY

Vice Overall Head, Entrepreneurship Cell, IIT Hyderabad

Core Member, Infero, Competitive Programming Club, IIT Hyderabad

Core Member, Epoch, Artificial Intelligence and Machine Learning Club, IIT Hyderabad

Core Member, Kludge, Cybersecurity & Networking Club, IIT Hyderabad

EXTRACURRICULAR

- Was one of four students selected from IIT Hyderabad for the **Sakura Science Program** held at the University of Tokyo.

- Won 2nd place in inter-hostel hackathon and 3rd place in inter-hostel capture-the-flag competitions.